G. Additional results on natural language benchmarks

We evaluate the models from Section 3.7 on standard natural language processing benchmarks: ARC Challenge (Yadav et al., 2019), COPA (Roemmele et al., 2011), Hellaswag (Zellers et al., 2019), Natural Questions (Kwiatkowski et al., 2019), PIQA (Bisk et al., 2019), SIQA (Sap et al., 2019) and TriviaQA (Joshi et al., 2017).

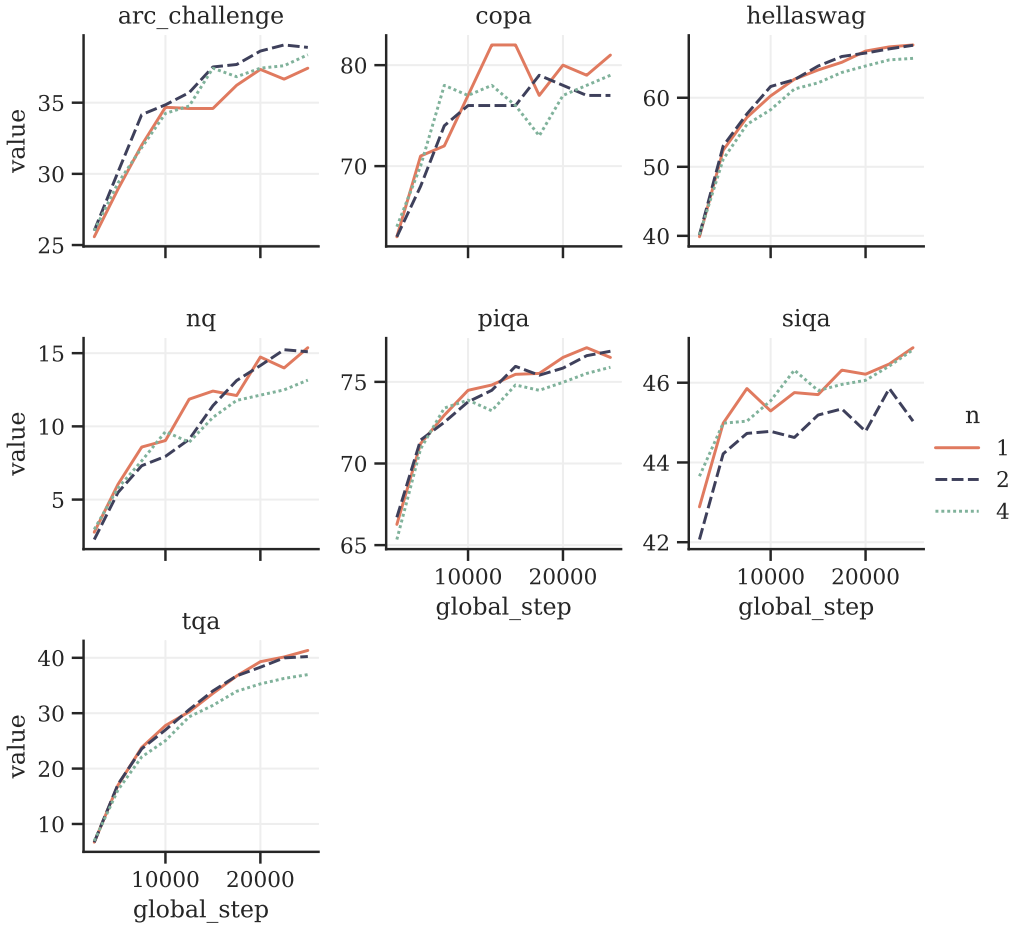


Figure S12: **Multiple token training with 7B models doesn't improve performance on choice tasks.** This figure shows the evolution of average accuracy of some standard NLP benchmarks (ARC Challenge COPA Hellaswag MMLU Natural Questions PIQA SIQA and TriviaQA. For the 7B models trained on 200B tokens of language data, the 2 future token model has the same performance as the baseline and the 4 future token model regresses a bit. Larger model sizes might be necessary to see improvements on these tasks.

H. Additional results on abstractive text summarization

In this section, we report comprehensive evaluation results on summarization tasks for the 7B parameter models trained on 200B and 500B tokens of natural language from Section 3.7.

Table S8: **Comprehensive evaluation on abstractive text summarization.** ROUGE-n (n-gram overlap) and ROUGE-L (longest common subsequence overlap) F_1 scores for 7B models trained on 200B and 500B tokens of natural language, respectively. The last three columns correspond to models trained on 500B tokens, the previous three to models trained on 200B tokens. Shown are numbers of the $n = 1$ baseline and the absolute difference of $n = 2$ and $n = 4$ models trained on the same number of tokens. Summary-level ROUGE-L ("ROUGE-L_{sum}") is reported where it differs from ROUGE-L. Model checkpoints with maximal validation ROUGE-L F_1 are selected separately for each model dataset and model type and reported in the first row corresponding to each dataset. Boldface for numbers within 0.05 difference to the best one for each dataset size separately.

Task	Metric	Baseline 200B	$\Delta_{n=2}$	$\Delta_{n=4}$	Baseline 500B	$\Delta_{n=2}$	$\Delta_{n=4}$
CNN/Dailymail (Nallapati et al., 2016)	*evaluation epoch*	2	2	2	2	2	2
	ROUGE-1	42.88	**+0.74**	**+0.74**	43.77	**+0.55**	**+0.50**
	ROUGE-2	19.56	**+0.52**	**+0.53**	20.34	**+0.52**	+0.34
	ROUGE-3	11.11	**+0.39**	**+0.35**	11.69	**+0.36**	+0.19
	ROUGE-L	29.72	**+0.66**	+0.49	30.51	**+0.48**	+0.37
	ROUGE-Lsum	40.18	**+0.72**	**+0.68**	41.02	**+0.56**	**+0.52**
Multi-News (Fabbri et al., 2019)	*evaluation epoch*	1	3	3	2	3	2
	ROUGE-1	44.48	**+1.70**	**+1.72**	45.87	**+1.05**	+0.69
	ROUGE-2	16.88	+0.44	**+0.70**	17.56	**+0.42**	**+0.40**
	ROUGE-3	9.63	-0.06	**+0.17**	9.91	**+0.22**	**+0.18**
	ROUGE-L	23.82	+0.17	**+0.40**	24.22	+0.20	**+0.26**
OrangeSum (Eddine et al., 2021)	*evaluation epoch*	2	2	3	2	1	3
	ROUGE-1	32.95	**+0.41**	+0.35	33.37	+0.32	**+0.78**
	ROUGE-2	13.90	**+0.31**	**+0.36**	14.22	+0.25	**+0.53**
	ROUGE-3	8.01	**+0.19**	**+0.21**	8.12	+0.22	**+0.48**
	ROUGE-L	23.62	+0.36	**+0.51**	23.91	+0.23	**+0.66**
pn-summary (Farahani et al., 2021)	*evaluation epoch*	1	1	1	1	2	3
	ROUGE-1	**1.03**	**+0.02**	**0.00**	0.92	**+0.09**	**+0.05**
	ROUGE-2	**0.13**	**+0.02**	**+0.03**	**0.15**	**0.00**	**0.00**
	ROUGE-3	**0.02**	**0.00**	**+0.02**	**0.02**	**0.00**	**+0.02**
	ROUGE-L	**1.02**	**+0.03**	**+0.01**	0.91	**+0.09**	**+0.05**
SAMSum (Gliwa et al., 2019)	*evaluation epoch*	3	3	3	3	3	3
	ROUGE-1	51.39	**+0.70**	+0.63	52.54	-0.24	**+0.69**
	ROUGE-2	26.46	**+0.76**	+0.30	27.74	-0.20	**+0.82**
	ROUGE-3	16.40	**+0.91**	+0.28	17.56	-0.30	**+0.71**
	ROUGE-L	42.59	**+0.90**	+0.51	43.92	-0.10	**+0.63**
ThaiSum (Chumpolsathien, 2020)	*evaluation epoch*	2	3	3	3	3	3
	ROUGE-1	45.08	+0.63	**+1.12**	45.48	+0.77	**+0.91**
	ROUGE-2	27.85	+0.30	**+0.73**	28.07	**+0.74**	+0.64
	ROUGE-3	15.73	+0.04	**+0.43**	15.82	**+0.50**	+0.30
	ROUGE-L	44.92	+0.64	**+1.12**	45.31	+0.76	**+0.89**
WikiSummary (Farahani, 2020)	*evaluation epoch*	3	3	3	3	3	3
	ROUGE-1	10.16	**+0.67**	-0.23	**12.80**	-0.17	-0.99
	ROUGE-2	**4.46**	-0.03	-0.09	**6.17**	-0.11	-0.69
	ROUGE-3	1.31	**+0.21**	+0.13	**1.98**	-0.08	-0.33
	ROUGE-L	10.11	**+0.65**	-0.28	**12.69**	-0.17	-0.99
XSum (Narayan et al., 2018)	*evaluation epoch*	2	2	3	2	2	3
	ROUGE-1	42.16	+0.71	**+1.07**	43.42	**+0.78**	+0.67
	ROUGE-2	19.19	**+0.54**	**+0.55**	20.32	**+0.68**	+0.34
	ROUGE-3	10.43	**+0.38**	+0.28	11.23	**+0.48**	+0.20
	ROUGE-L	34.03	+0.67	**+0.92**	35.18	**+0.79**	+0.63

Table S9: **Performance on abstractive text summarization.** ROUGE-L (longest common subsequence overlap) F_1 score for 7B models trained on 200B and 500B tokens of natural language. We finetune the respective models on each task's training data separately for a given number of epochs and select the checkpoints with maximal ROUGE-L F_1 on the validation dataset. The second and fifth column report the numbers for a next-token prediction model, while the third, fourth, sixth and seventh one report the absolute improvements for 2-token and 4-token prediction models trained on the same amount of data, respectively. Boldface for numbers within 0.05 difference to the best one for each dataset size separately.

Dataset	Baseline 200B	$\Delta_{n=2}$	$\Delta_{n=4}$	Baseline 500B	$\Delta_{n=2}$	$\Delta_{n=4}$
CNN/Dailymail	29.72	**+0.66**	+0.49	30.51	**+0.48**	+0.37
Multi-News	23.82	+0.17	**+0.40**	24.22	+0.20	**+0.26**
OrangeSum	23.62	+0.36	**+0.51**	23.91	+0.23	**+0.66**
pn-summary	**1.02**	**+0.03**	**+0.01**	0.91	**+0.09**	**+0.05**
SAMSum	42.59	**+0.90**	+0.51	43.92	-0.10	**+0.63**
ThaiSum	44.92	+0.64	**+1.12**	45.31	+0.76	**+0.89**
WikiSummary	10.11	**+0.65**	-0.28	**12.69**	-0.17	-0.99
XSum	34.03	+0.67	**+0.92**	35.18	**+0.79**	+0.63
Average	26.23	**+0.51**	**+0.46**	27.08	**+0.28**	**+0.31**

Table S10: **Summary statistics for abstractive text summarization evaluations.** Reported are averages for ROUGE-n and ROUGE-L metrics across all datasets from Table S8, separately for precision, recall and F_1 score. Both 2-token and 4-token prediction models outperform the next-token prediction baseline. Trained on 500B tokens, 4-token prediction models appear better at recall metrics while 2-token prediction models appear better at precision metrics. Model checkpoints are selected as described in Table S8. Boldface for numbers within 0.05 difference to the best one for each dataset size separately.

Metric	Aspect	Baseline 200B	$\Delta_{n=2}$	$\Delta_{n=4}$	Baseline 500B	$\Delta_{n=2}$	$\Delta_{n=4}$
ROUGE-1	F_1	33.77	**+0.70**	**+0.68**	34.77	**+0.39**	**+0.41**
	precision	35.76	**+0.88**	**+0.83**	37.03	**+0.42**	-0.04
	recall	34.37	**+0.45**	**+0.45**	35.14	+0.35	**+0.68**
ROUGE-2	F_1	16.06	**+0.36**	**+0.39**	16.82	**+0.29**	**+0.30**
	precision	16.97	**+0.40**	**+0.43**	17.91	**+0.29**	+0.03
	recall	16.34	+0.28	**+0.35**	16.99	+0.32	**+0.48**
ROUGE-3	F_1	9.08	**+0.26**	**+0.23**	9.54	**+0.18**	**+0.22**
	precision	9.59	**+0.29**	**+0.28**	10.17	**+0.18**	+0.05
	recall	9.26	**+0.21**	**+0.20**	9.65	+0.21	**+0.35**
ROUGE-L	F_1	26.23	**+0.51**	**+0.46**	27.08	**+0.28**	**+0.31**
	precision	27.79	**+0.62**	+0.55	28.85	**+0.28**	-0.09
	recall	26.71	**+0.37**	**+0.32**	27.40	+0.28	**+0.57**
ROUGE-L_{sum}	F_1	27.53	**+0.52**	**+0.48**	28.40	**+0.29**	**+0.33**
	precision	29.07	**+0.64**	+0.58	30.15	**+0.29**	-0.08
	recall	28.13	**+0.35**	**+0.33**	28.81	+0.29	**+0.60**

I. Additional results on mathematical reasoning in natural language

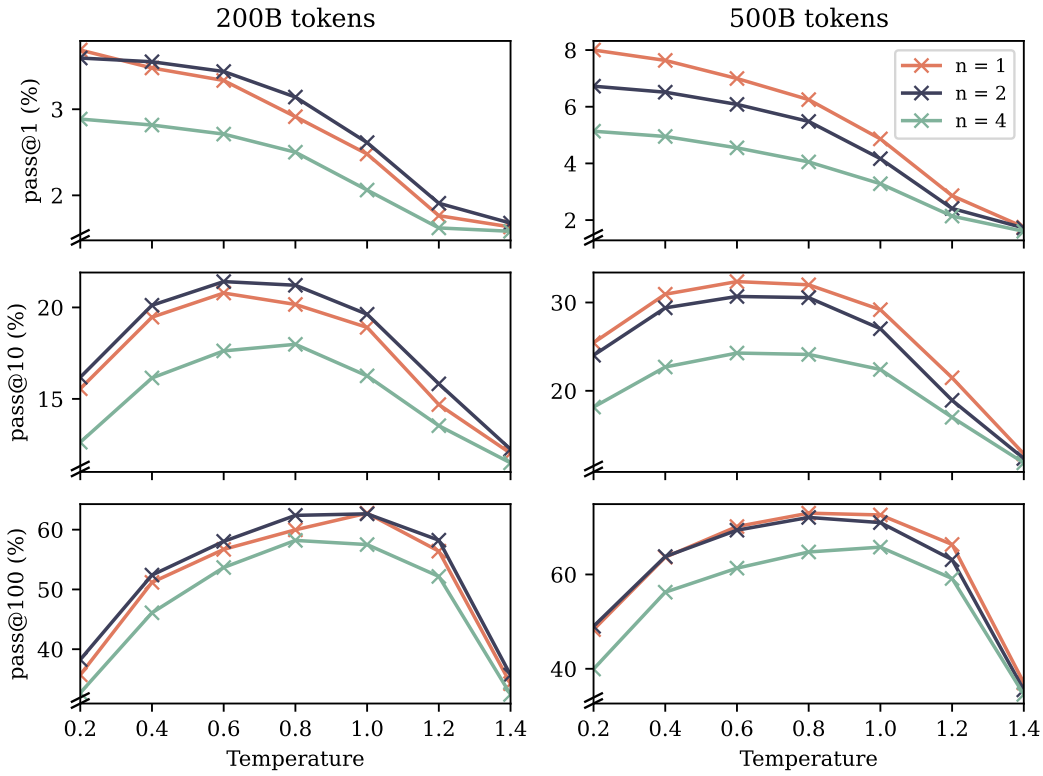


Figure S13: **Performance on the mathematical reasoning benchmark GSM8K (Cobbe et al., 2021).** We evaluate pretrained next-token and multi-token prediction models trained on 200B and 500B tokens of natural language in 8-shot mode using nucleus sampling (Holtzman et al., 2020) with probability mass 0.95 and various sampling temperatures. Reported are the frequencies of the correct final answer to appear among k samples, for $k = 1, 10, 100$, estimated from 200 samples like in code generation benchmarks (Chen et al., 2021). After 200B tokens, the 2-token prediction model has a clear advantage over the next-token baseline but the order reverses after 500B tokens. The 4-token prediction model is worse throughout. We interpret this similarly to the findings in Section 4.1: the follow-your-nose chains-of-thought required for GSM8K may be difficult to learn from a limited amount of data, attesting to the data efficiency of multi-token prediction training. Once the correct circuits for correct autoregressive chains-of-thought in this domain have formed, however, multi-token prediction comes at a cost.